

Formulario: Electrodinámica Avanzada

ONDAS PLANAS Y POLARIZACIÓN

$$\mathbf{E}(\mathbf{r}, t) = \text{Re} \left[\mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \right], \quad \mathbf{B}(\mathbf{r}, t) = \text{Re} \left[\mathbf{B}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \right]$$

$$\nabla^2 \mathbf{E} - \mu \epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0, \quad \nabla^2 \mathbf{H} - \mu \epsilon \frac{\partial^2 \mathbf{H}}{\partial t^2} = 0$$

$$v = \frac{1}{\sqrt{\mu \epsilon}}, \quad n = \frac{c}{v} = \sqrt{\frac{\epsilon \mu}{\epsilon_0 \mu_0}}, \quad Z = \sqrt{\frac{\mu}{\epsilon}} = \frac{Z_0}{n}$$

$$\mathbf{k} \cdot \mathbf{E}_0 = 0, \quad \mathbf{k} \cdot \mathbf{B}_0 = 0, \quad k^2 = \mu \epsilon \omega^2$$

$$\mathbf{B}_0 = \frac{1}{\omega} \mathbf{k} \times \mathbf{E}_0, \quad \mathbf{H}_0 = \frac{1}{Z} \hat{\mathbf{k}} \times \mathbf{E}_0$$

$$\langle u \rangle = \frac{1}{4} \epsilon |\mathbf{E}_0|^2 + \frac{1}{4\mu} |\mathbf{B}_0|^2 = \frac{1}{2} \epsilon |\mathbf{E}_0|^2$$

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}, \quad \langle \mathbf{S} \rangle = \frac{1}{2} \text{Re}(\mathbf{E}_0 \times \mathbf{H}_0^*) = \frac{|\mathbf{E}_0|^2}{2Z} \hat{\mathbf{k}}$$

$$\mathbf{J} = \begin{pmatrix} E_x \\ E_y \end{pmatrix} = \begin{pmatrix} a_x e^{i\delta_x} \\ a_y e^{i\delta_y} \end{pmatrix}, \quad \Delta = \delta_y - \delta_x$$

$$\frac{E_x^2}{a_x^2} + \frac{E_y^2}{a_y^2} - 2 \frac{E_x E_y}{a_x a_y} \cos \Delta = \sin^2 \Delta$$

$$S_0 = a_x^2 + a_y^2, \quad S_1 = a_x^2 - a_y^2$$

$$S_2 = 2a_x a_y \cos \Delta, \quad S_3 = 2a_x a_y \sin \Delta$$

$$S_0^2 \geq S_1^2 + S_2^2 + S_3^2 \quad (\text{Esfera de Poincaré})$$

INTERFACES Y COEFICIENTES DE FRESNEL

$$\theta_r = \theta_i, \quad n_1 \sin \theta_i = n_2 \sin \theta_t \quad (\text{Snell})$$

$$r_s = \frac{n_1 \cos \theta_i - n_2 \cos \theta_t}{n_1 \cos \theta_i + n_2 \cos \theta_t}, \quad t_s = \frac{2n_1 \cos \theta_i}{n_1 \cos \theta_i + n_2 \cos \theta_t}$$

$$r_p = \frac{n_2 \cos \theta_i - n_1 \cos \theta_t}{n_2 \cos \theta_i + n_1 \cos \theta_t}, \quad t_p = \frac{2n_1 \cos \theta_i}{n_2 \cos \theta_i + n_1 \cos \theta_t}$$

$$R_{s,p} = |r_{s,p}|^2, \quad T_{s,p} = \frac{n_2 \cos \theta_t}{n_1 \cos \theta_i} |t_{s,p}|^2, \quad R + T = 1$$

$$\tan \theta_B = \frac{n_2}{n_1} \implies r_p = 0 \quad (\text{Ángulo de Brewster})$$

$$\theta_i > \theta_c = \arcsin \left(\frac{n_2}{n_1} \right), \quad \kappa = \frac{\omega}{c} \sqrt{n_1^2 \sin^2 \theta_i - n_2^2}$$

MEDIOS DISPERSIVOS, CONDUCTORES Y PLASMAS

$$\nabla^2 \mathbf{E} - \mu \epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} - \mu \sigma \frac{\partial \mathbf{E}}{\partial t} = 0 \quad (\mathbf{J}_f = \sigma \mathbf{E})$$

$$\epsilon_c(\omega) = \epsilon + i \frac{\sigma}{\omega}, \quad k_c^2 = \mu \epsilon_c(\omega) \omega^2, \quad k_c = \beta + i \frac{\alpha}{2}$$

$$\sigma \gg \omega \epsilon \implies \beta \approx \frac{\alpha}{2} \approx \sqrt{\frac{\mu \sigma \omega}{2}}, \quad \delta = \frac{2}{\alpha} = \sqrt{\frac{2}{\mu \sigma \omega}}$$

$$\omega_p = \sqrt{\frac{N e^2}{m \epsilon_0}}, \quad \epsilon(\omega) = \epsilon_0 \left(1 - \frac{\omega_p^2}{\omega^2} \right), \quad \omega^2 = \omega_p^2 + c^2 k^2$$

$$v_\phi = \frac{\omega}{k} = \frac{c}{\sqrt{1 - \omega_p^2/\omega^2}}, \quad v_g = \frac{d\omega}{dk} = c \sqrt{1 - \frac{\omega_p^2}{\omega^2}}$$

$$v_\phi v_g = c^2, \quad k(\omega) \approx k(\omega_0) + \frac{1}{v_g}(\omega - \omega_0) + \frac{1}{2} k''(\omega_0)(\omega - \omega_0)^2$$

CAUSALIDAD Y KRAMERS-KRONIG

$$\mathbf{P}(t) = \epsilon_0 \int_{-\infty}^t \chi(t-t') \mathbf{E}(t') dt' \implies \chi(-\omega) = \chi^*(\omega)$$

$$\text{Re} \chi(\omega) = \frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} \frac{\text{Im} \chi(\omega')}{\omega' - \omega} d\omega'$$

$$\text{Im} \chi(\omega) = -\frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} \frac{\text{Re} \chi(\omega')}{\omega' - \omega} d\omega'$$

$$\text{Re} \epsilon(\omega) - 1 = \frac{2}{\pi} \mathcal{P} \int_0^{\infty} \frac{\omega' \text{Im} \epsilon(\omega')}{\omega'^2 - \omega^2} d\omega'$$

$$\text{Im} \epsilon(\omega) = -\frac{2\omega}{\pi} \mathcal{P} \int_0^{\infty} \frac{\text{Re} \epsilon(\omega') - 1}{\omega'^2 - \omega^2} d\omega'$$

$$\epsilon(\omega) = 1 + \int_0^{\infty} \frac{f(\Omega)}{\Omega^2 - \omega^2 - i\Gamma\omega} d\Omega$$

$$\lim_{\Gamma \rightarrow 0^+} \text{Im} \epsilon(\omega) = \frac{\pi}{2\omega} f(\omega) \quad (\omega > 0)$$

GUÍAS DE ONDA

$$\mathbf{E} = [\mathbf{E}_\perp + E_z \hat{\mathbf{z}}] e^{i(kz - \omega t)}, \quad \mathbf{B} = [\mathbf{B}_\perp + B_z \hat{\mathbf{z}}] e^{i(kz - \omega t)}$$

$$(\nabla_\perp^2 + \gamma^2) E_z = 0, \quad (\nabla_\perp^2 + \gamma^2) B_z = 0, \quad \gamma^2 = \frac{\omega^2}{c^2} - k^2$$

$$\mathbf{E}_\perp = \frac{i}{\gamma^2} (k \nabla_\perp E_z - \omega \hat{\mathbf{z}} \times \nabla_\perp B_z)$$

$$\mathbf{B}_\perp = \frac{i}{\gamma^2} \left(k \nabla_\perp B_z + \frac{\omega}{c^2} \hat{\mathbf{z}} \times \nabla_\perp E_z \right)$$

$$\text{TM: } B_z = 0, \quad E_z|_S = 0; \quad \text{TE: } E_z = 0, \quad \frac{\partial B_z}{\partial n} \Big|_S = 0$$

$$\text{Guía Rec: } \omega_{mn} = c \gamma_{mn} = c \sqrt{\left(\frac{m\pi}{a} \right)^2 + \left(\frac{n\pi}{b} \right)^2}$$

$$k_{mn}(\omega) = \frac{1}{c} \sqrt{\omega^2 - \omega_{mn}^2}, \quad v_\phi v_g = c^2$$

$$Z_{\text{TE}} = \frac{Z_0}{\sqrt{1 - (\omega_{mn}/\omega)^2}}, \quad Z_{\text{TM}} = Z_0 \sqrt{1 - \left(\frac{\omega_{mn}}{\omega} \right)^2}$$

$$\alpha = \frac{P_{\text{dis}}}{2P_{\text{trans}}}, \quad P_{\text{dis}} = \frac{1}{2\sigma\delta} \oint_S |\mathbf{H}_{\text{tang}}|^2 dl$$

POTENCIALES RETARDADOS Y RADIACIÓN

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \mathbf{J}(\mathbf{r}') \frac{e^{ik|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|} d^3 r'$$

$$r \gg \lambda \implies \mathbf{A}(\mathbf{r}) \approx \frac{\mu_0}{4\pi} \frac{e^{ikr}}{r} \int \mathbf{J}(\mathbf{r}') e^{-ik\hat{\mathbf{r}} \cdot \mathbf{r}'} d^3 r'$$

$$\mathbf{B} \approx ik\hat{\mathbf{r}} \times \mathbf{A}, \quad \mathbf{E} \approx c\mathbf{B} \times \hat{\mathbf{r}}$$

$$\mathbf{p} = \int \mathbf{r}' \rho(\mathbf{r}') d^3 r', \quad \mathbf{A}(\mathbf{r}) = -\frac{i\mu_0 \omega}{4\pi} \frac{\mathbf{p} e^{ikr}}{r}$$

$$\mathbf{B} = \frac{\mu_0 \omega^2}{4\pi c} (\hat{\mathbf{r}} \times \mathbf{p}) \frac{e^{ikr}}{r}, \quad \mathbf{E} = \frac{\mu_0 \omega^2}{4\pi} [(\hat{\mathbf{r}} \times \mathbf{p}) \times \hat{\mathbf{r}}] \frac{e^{ikr}}{r}$$

$$\frac{dP}{d\Omega} = \frac{\mu_0 \omega^4}{32\pi^2 c} |\mathbf{p}|^2 \sin^2 \theta, \quad P_{\text{total}} = \frac{\mu_0 \omega^4}{12\pi c} |\mathbf{p}|^2$$

$$R_{\text{rad}} = \frac{2P_{\text{total}}}{I_0^2} = \frac{\mu_0 \omega^2 d^2}{6\pi c} \quad (\text{Dipolo eléctrico } E1)$$

ELECTRODINÁMICA COVARIANTE

$$\eta_{\mu\nu} = \eta^{\mu\nu} = \text{diag}(1, -1, -1, -1)$$

$$x^\mu = (ct, \mathbf{r}), \quad \partial_\mu = \left(\frac{1}{c} \frac{\partial}{\partial t}, \nabla \right), \quad \square = \partial_\mu \partial^\mu = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2$$

$$U^\mu = \gamma(c, \mathbf{v}), \quad p^\mu = mU^\mu = (E/c, \mathbf{p}), \quad p_\mu p^\mu = m^2 c^2$$

$$J^\mu = (c\rho, \mathbf{J}), \quad A^\mu = (\Phi/c, \mathbf{A}), \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix}$$

$$\tilde{F}^{\mu\nu} = \frac{1}{2} \epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}, \quad (\mathbf{E}/c \rightarrow \mathbf{B}, \quad \mathbf{B} \rightarrow -\mathbf{E}/c)$$

$$\partial_\mu F^{\mu\nu} = \mu_0 J^\nu, \quad \partial_\mu \tilde{F}^{\mu\nu} = 0, \quad \partial_\mu J^\mu = 0$$

$$\partial_\mu A^\mu = 0 \implies \square A^\mu = \mu_0 J^\mu \quad (\text{Gauge de Lorenz})$$

$$F_{\mu\nu} F^{\mu\nu} = 2 \left(B^2 - \frac{E^2}{c^2} \right), \quad F_{\mu\nu} \tilde{F}^{\mu\nu} = -\frac{4}{c} (\mathbf{E} \cdot \mathbf{B})$$

DINÁMICA RELATIVISTA Y DRIFTS

$$E'_x = E_x, \quad B'_x = B_x$$

$$E'_y = \gamma(E_y - vB_z), \quad B'_y = \gamma \left(B_y + \frac{v}{c^2} E_z \right)$$

$$E'_z = \gamma(E_z + vB_y), \quad B'_z = \gamma \left(B_z - \frac{v}{c^2} E_y \right)$$

$$K^\mu = \frac{dp^\mu}{d\tau} = qF^{\mu\nu} U_\nu \implies \frac{d\mathbf{p}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

$$L = -mc^2 \sqrt{1 - \frac{v^2}{c^2}} - q\Phi + q\mathbf{A} \cdot \mathbf{v}$$

$$\mathbf{P}_{\text{can}} = \mathbf{p} + q\mathbf{A}, \quad H = \sqrt{(\mathbf{P} - q\mathbf{A})^2 c^2 + m^2 c^4} + q\Phi$$

$$\mathbf{v}_E = \frac{\mathbf{E} \times \mathbf{B}}{B^2}, \quad \mu = \frac{mv_\perp^2}{2B} = \text{const} \implies \frac{B_{\text{esp}}}{B_0} = \frac{v_0^2}{v_{\perp 0}^2}$$

ACCIÓN Y TENSORES DE ENERGÍA

$$\mathcal{L}_{\text{Maxwell}} = -\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} - J_\mu A^\mu$$

$$\mathcal{L}_{\text{Proca}} = -\frac{1}{4\mu_0} F_{\mu\nu} F^{\mu\nu} + \frac{\mu_\gamma^2}{2\mu_0} A_\mu A^\mu \quad (\text{Masa fotón})$$

$$T_{\text{can}}^{\mu\nu} = -\frac{1}{4\mu_0} \eta^{\mu\nu} F_{\alpha\beta} F^{\alpha\beta} - \frac{1}{\mu_0} F^{\mu\rho} \partial^\nu A_\rho$$

$$\Theta^{\mu\nu} = T_{\text{can}}^{\mu\nu} + \frac{1}{\mu_0} F^{\mu\rho} \partial_\rho A^\nu = \frac{1}{\mu_0} \left(F^{\mu\rho} F^\nu{}_\rho + \frac{1}{4} \eta^{\mu\nu} F_{\alpha\beta} F^{\alpha\beta} \right)$$